Table 10
Equipment installed cost breakdown for base case, multi-stage CO_2-FTS and CO_2-FTS with tail gas recycle.

	Base case	Multi-stage CO_2-FTS		CO_2-FTS with tail gas recycle			
		Two-stage	Three-stage	63.8 % recycle	80.3 % recycle	90 % recycle	
Equipment	Equipment installed cost (M$)	Equipment installed cost (M$)	Equipment installed cost (M$)	Equipment installed cost (M$)	Equipment installed cost (M$)	Equipment installed cost (M$)	
H_2 compressor	57.61	57.61	57.61	57.61	57.61	57.61	
CO_2 compressor	13.14	13.14	13.14	13.14	13.14	13.14	
Heater	1.87	3.52	4.54	2.92	3.37	3.47	
CO_2-FTS reactor	55.25	93.33	119.11	82.72	95.49	104.55	
Cooler	0.23	0.41	0.56	0.29	0.31	0.32	
Three-phase separator	0.17	0.31	0.43	0.18	0.19	0.19	

flowsheets are shown in Figs. 5–7. APEA calculates the purchased and installed equipment cost for CO_2-FTS process such as the compressors, heater, cooler, from which the total direct cost of the plant is calculated. CAPEX is derived from the total direct cost with the accounting of other financial items such as indirect cost, engineering and contingency. The OPEX consists of the fixed operating and maintenance costs (fixed OPEX) and the variable operating and maintenance costs (variable OPEX). The fixed OPEX assumed as 3 % of CAPEX is made up of the total maintenance cost, labour costs, administration cost, and other cost [34]. The variable OPEX is based on feedstock cost and utility cost obtained from the APEA with their unit price.

The financial parameters used in the economic evaluation are shown in Table 9. CO_2 is assumed to be free from a CO_2 capture plant [35]. The hydrogen price was assumed to be $2/kg which is the 2025 target of renewable hydrogen from electrolysis [36]. Other parameters are commonly used for the economic evaluation of a commercial plant [25,34,37].

It is noted that the CO_2-FTS reactor cannot be mapped in APEA, therefore it was considered as zero cost. Alternatively, Eqs. (17) and (18) are used to calculate the equipment installed cost of the CO_2-FTS reactor, using an installation factor of 2.75 and a scaling exponent of 0.8. The base cost and base size are referred to Zang et al. [24].

$$Equipment\ installed\ cost = Purchased\ equipment\ cost \times Installation\ factor \tag{17}$$

$$Purchased\ equipment\ cost = Base\ cost \times \left(\frac{New\ size}{Base\ size}\right)^{Scaling\ exponent} \tag{18}$$

The annual capital cost (ACC) was calculated by annualizing the CAPEX using Eq. (19) with plant lifetime n, and interest rate i.

$$ACC = CAPEX\left(\frac{i(1+i)^n}{(1+i)^n - 1}\right) \tag{19}$$

The total annual cost shown in Eq. (20) is the sum of ACC and OPEX.

$$Total\ annual\ cost = ACC + Fixed\ OPEX + Variable\ OPEX \tag{20}$$

The MSP of jet fuel was obtained from the Eq. (21) [38].

$$MSP = \frac{Total\ annual\ cost}{Operating\ time\ per\ year \times Y_{Jetfuel}} \tag{21}$$

5.2. Economic evaluation results

The economic performance of the CO_2-FTS process was performed for base case, multi-stage CO_2-FTS, and CO_2-FTS with tail gas recycle process. The installed equipment costs in Table 10 were obtained from APEA. It is noticeable that compressors and CO_2-FTS reactors dominate the total equipment installed cost for different CO_2-FTS processes. The installed cost of the CO_2 compressor and H_2 compressor remain constant at 13.1 M$ and 57.6 M$ for all CO_2-FTS processes because the feedstock is only compressed initially and there is no pressure drop after gas compression. While the increase in equipment installed cost of the CO_2-FTS reactor can be seen for both water removal configurations. With the addition of more stages, the installed cost of the CO_2-FTS reactor is 93.3 M$ for two stages process and 119.1 M$ for three stages process. The increase in these costs is due to the addition of more reactors in the multi-stage process. For the tail gas recycle process, as the recycle ratio changes from 0 to 90 %, the installed cost of the CO_2-FTS reactor varies from 55.3 M$ to 104.6 M$ due to the larger reactor size required.

Table 11 presents the CAPEX breakdown and Table 12 summarises the economic performance. The best economic performance for jet fuel production is achieved by CO_2-FTS with 90 % tail gas recycle. This process can potentially reduce the MSP from $5.87/kg to $2.6/kg, which is only 44 % of the base case price. For comparison with current market